# Project Proposal Report: COVID-19 Data Analysis

## 1) Dataset

**Title:** Our World in Data — COVID-19 Dataset
**Area:** Global Public Health & Epidemiology
**Size:** ~350,000+ rows × 67 columns
**Source:** https://ourworldindata.org/covid-deaths

**References (Previous Works Using Dataset):**

- Our World in Data COVID-19 Research
- WHO COVID-19 Dashboard
- Various academic studies on COVID-19 trends, vaccination impact, and mortality analysis

This dataset contains global daily records of COVID-19 cases, deaths, testing, vaccination, and socio-economic indicators for around 180+ countries.

## 2) Attributes (Type/Category of Records)

The dataset consists of two main types of attributes:

**Numerical Attributes (Continuous):**

- total_cases, new_cases, total_deaths
- people_vaccinated, total_tests
- gdp_per_capita, life_expectancy
- hospital_beds_per_thousand, median_age

**Categorical Attributes:**

- iso_code (country code)
- continent
- location (country name)
- date
- tests_units

# 3) Analysis & Tools

**Possible Analysis:**

- Exploratory Data Analysis (EDA)
- Missing value analysis and handling
- Trend analysis (cases, deaths, vaccinations over time)
- Correlation analysis between variables
- Statistical tests (ANOVA, Chi-square)
- Predictive modeling (Regression, Classification)

**Tools & Technologies:**

- Python
- Pandas, NumPy (data processing)
- Matplotlib, Seaborn (visualization)
- SciPy (statistical analysis)
- Scikit-learn (machine learning models)
- Google Colab (development platform)

# 4) Applications

- COVID-19 Monitoring Dashboard
- Death rate prediction system using Machine Learning
- Vaccination analysis and policy support tools
- Public health decision support system
- Country risk classification (high-risk vs low-risk)
- Research and educational data analysis platform

# Conclusion

This project aims to analyze global COVID-19 data to extract meaningful insights and build predictive models. The dataset is rich and suitable for data analysis, visualization, and machine learning applications, making it highly relevant for real-world problem solving.